

Title: SmartSales – Sales and Inventory Analytics App

Android Application Project Proposal

Introduction

SmartSales is an Android mobile application built with Kotlin that helps small businesses manage their products, stock, customers, and sales transactions in one place, while turning that day-to-day data into clear business analytics. Instead of tracking sales and inventory in notebooks or spreadsheets, business owners and their sales staff can record every transaction directly in the app and instantly see how the business is performing. Firebase Authentication provides secure, role-based login for Admin and Sales Staff accounts, while Firebase Realtime Database stores and synchronizes product, stock, customer, and sales data in real time.

Background Study

Small businesses and student-run shops commonly record sales and stock levels manually, which makes it slow and error-prone to answer basic operational questions such as which products are selling best, how much revenue was generated this month, or which items are about to run out of stock. Existing solutions are often either too simple – offering only basic CRUD-style record keeping with no insight – or too complex, aimed at large enterprises with full ERP systems that are impractical for a small business or a student project to adopt.

SmartSales addresses this gap by combining a straightforward sales and inventory workflow with a built-in analytics dashboard. As products are added, stock is adjusted, and sales are recorded, the application automatically computes key business metrics – total revenue, total sales, average order value, top-selling products, category performance, and low-stock alerts – so that business owners can make informed decisions without needing separate reporting tools. This project also demonstrates a complete data pipeline suitable for a Data Analyst-oriented academic portfolio: data capture, storage, filtering, aggregation, and visualization.

Requirements

- **User Authentication:** New users shall be able to register and existing users shall be able to log in using email/password via Firebase Authentication. The app shall support two separate user roles – Admin and Sales Staff – each with its own account credentials. The system shall automatically route each user to the correct dashboard based on their stored role; role switching within the app shall not be permitted.
- **Role-Based Access Control:** Role permissions shall be enforced both in the app interface and at the database/security-rule level, so that a Sales Staff account cannot perform Admin-only actions (such as adding stock or deleting products) even through a direct database request.

- **Product and Category Management:** Admin users shall be able to add, edit, delete, search, and filter products, and to organize products into categories. Each product record shall include name, category, price, stock quantity, supplier, and an optional product image.
- **Inventory Management:** The system shall track stock levels for every product. Admin users shall be able to add stock manually (restocking), and stock shall be reduced automatically whenever a sale is completed. Stock quantity shall never be allowed to fall below zero, and each stock update shall be applied exactly once per transaction.
- **Customer Management:** The system shall allow customer records (name, phone number, and optional email) to be created and searched during the sales process, and shall allow Admin users to edit customer details.
- **Sales Transactions:** Both Admin and Sales Staff shall be able to record a sale by selecting a product, entering a valid quantity, and optionally associating a customer. Each sale record shall store the product, quantity, total amount, date/time, and the identity of the staff member who made the sale. Quantity shall be required, greater than zero, and shall not exceed the currently available stock.
- **Sales History and Filtering:** Admin users shall be able to view the complete sales history for the business, while Sales Staff shall be able to view their own recorded sales. Sales history shall be filterable by date range, product, and category.
- **Data Analytics Dashboard:** The system shall calculate and display key business metrics, including total revenue, total sales, total products, low-stock products, total customers, average order value, top-selling product, and top-performing category, along with supporting charts such as monthly sales trends, sales by category, and stock status breakdown.
- **Data Access and Security:** All data operations require Internet connectivity to synchronize with Firebase. Firebase Security Rules shall ensure that each user can only access data and functions appropriate to their role. The app must support a minimum SDK suitable for current-generation Android devices.

Features

- **Secure Role-Based Login:** Separate Admin and Sales Staff accounts using Firebase Authentication, with automatic dashboard routing based on the account's stored role and no manual role switching.
- **Product & Category Management:** Admin can add, edit, delete, search, and filter products and categories, with optional product images stored in Firebase Storage.
- **Inventory & Stock Control:** Admin-only Add Stock function, automatic and transaction-safe stock deduction after every sale, and clear In Stock / Low Stock / Out of Stock status indicators.
- **Sales Entry:** A guided flow – Select Product → Enter Quantity → Select Customer → Review → Save – available to both Admin and Sales Staff, with built-in quantity and stock validation.
- **Customer Records:** Quick customer lookup and creation during the sales flow to avoid duplicate entries.
- **Sales History:** Chronological sales history with date, product, and category filters, plus an itemized sale detail/receipt view.

- **Analytics Dashboard:** Real-time KPI cards and charts (monthly sales, sales by category, stock status) that recompute automatically as new data is recorded, giving business owners an at-a-glance view of performance.
- **Stable, Crash-Resistant Operations:** Safe handling of missing or unexpected product data, validated user input, and transaction-based stock updates so that inventory and sales operations never corrupt data or crash the application.

Tools

- **Android Studio & Kotlin:** The project will be developed using Kotlin, the recommended language for modern Android development, following an MVVM architecture that separates UI, state, and data layers.
- **Firebase Realtime Database:** Product, category, customer, and sales data will be stored as JSON in the cloud with real-time synchronization, so that Admin and Sales Staff always see consistent, up-to-date stock and sales information.
- **Firebase Authentication:** Used to manage secure, role-based email/password login and registration for Admin and Sales Staff accounts.
- **Firebase Storage:** Used to store optional product images referenced by product records.
- **Android Jetpack Libraries:** Standard AndroidX components (ViewModel, LiveData/StateFlow) and Material Design libraries will be used for the UI, along with a charting library (such as MPAndroidChart) for the analytics dashboard.
- **Other Tools:** Gradle will be used for dependency management (including Firebase configuration). Testing will be carried out on Android emulators and physical devices, covering role-based access, stock and sales calculations, and edge cases such as missing product data.

Future Scope

Looking ahead, SmartSales can be extended in several directions. Online/in-app payment integration could allow customers to pay digitally at the point of sale. An AI-based forecasting module could predict future demand and suggest optimal restocking levels based on historical sales trends. The system could also be expanded to support multiple branches or warehouses, integrate with external accounting or ERP platforms, and provide offline-first functionality with automatic conflict resolution once connectivity is restored. Additional enhancements might include supplier management, purchase orders, barcode/QR scanning for faster product entry, customer loyalty tracking, and exportable business reports. Through these extensions, SmartSales can evolve from a student project into a comprehensive small-business management platform.

Conclusion

The SmartSales application aims to give small business owners a simple, reliable way to manage products, inventory, customers, and sales, while automatically transforming that data into actionable business insight. By combining secure role-based authentication, a complete sales and inventory

workflow, and a real-time analytics dashboard, the app reduces the manual effort of tracking business performance and helps owners make faster, better-informed decisions. The initial implementation demonstrates the feasibility of a lightweight, analytics-driven business management system built entirely on modern Android and Firebase technologies.